

106G NRZ CPO TX Budget — OCI Gen2

System: Lightmatter OCI Gen2, GlobalFoundries process
Modulation: NRZ, 106.25 GBaud per lane, 4λ DWDM
Date: 2026-08-02
Sources: 200G-OCI Optical Phy Spec v1.0 (Gen1 baseline), IEEE 802.3ck-2022, IEEE P802.3dj D1.3, ECEN721 Lecture 7 (TX Analysis), ECEN720 Lecture 10 (Jitter)

1. Overview and Scope

This document builds the transmitter-side link budget for OCI Gen2 at 106.25 GBaud NRZ. The OCI Gen1 spec (53.125 GBaud) is the normative baseline; all per-UI timing parameters scale by ÷2, all frequency/bandwidth parameters scale by ×2, and all optical power parameters are unchanged. The TX chain runs from the External Laser Source (ELS) through the OCI chiplet's MRM driver → Micro-Ring Resonator (MRM) modulator to the fiber attach point (TP2).

BER Conventions in This Document

Two distinct BER thresholds are used — they apply to different layers:

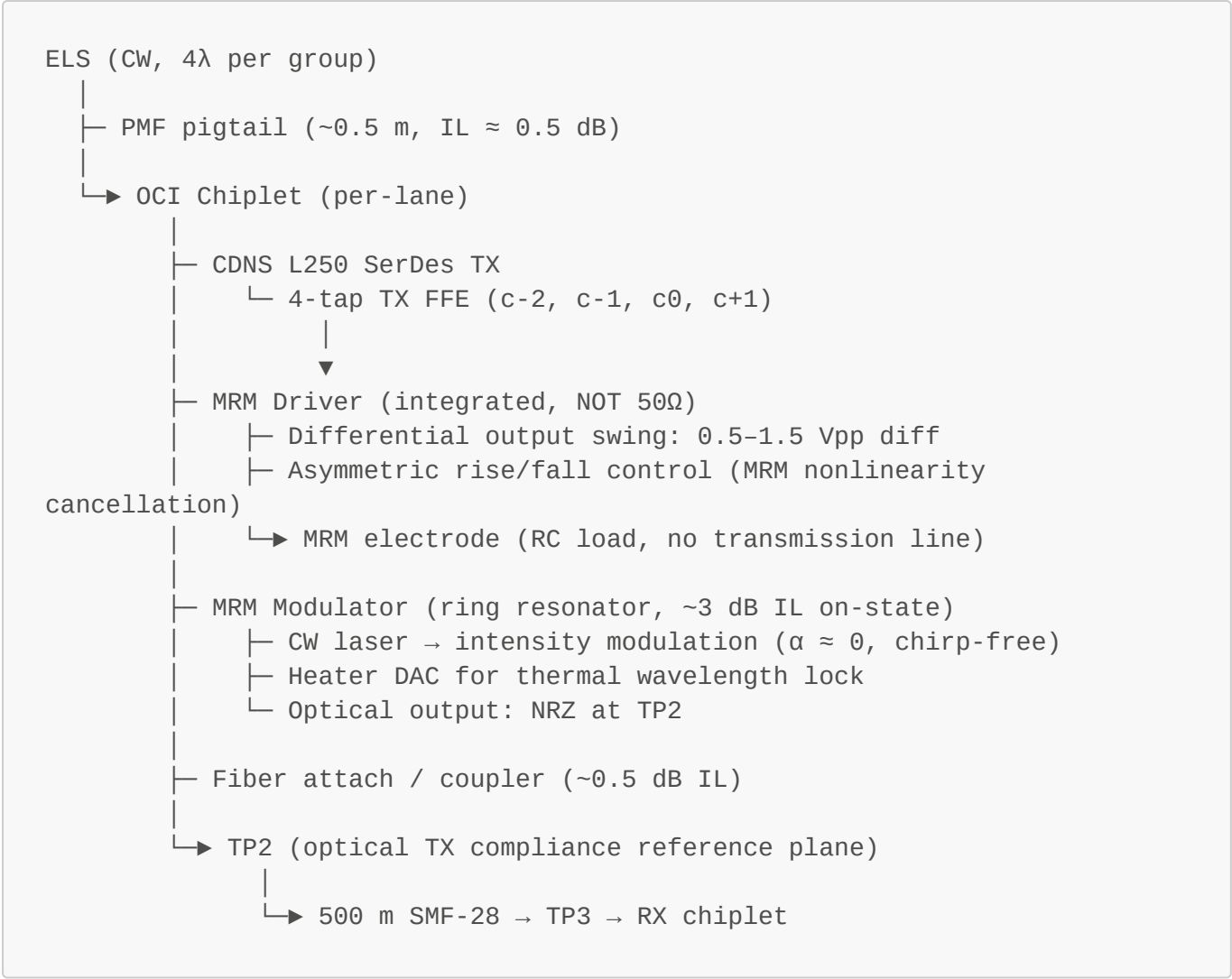
Layer	BER target	Q	Applies to
TX electrical driver jitter	1×10^{-12}	7.035	Driver DJ/RJ/PSIJ spec, eye mask
OCI optical TDEC measurement	2.4×10^{-4}	3.54	Optical TX compliance (OCI Gen2 spec)

The 1e-12 BER is the uncoded raw link BER target (conservative; no FEC assumed in the driver spec). The 2.4e-4 threshold is the KP4 pre-FEC BER used in the OCI optical TDEC compliance measurement only.

Scope

- TX driver electrical specification (swing, EQ, rise/fall, jitter, eye mask)
- OCI Gen2 TX optical specifications (scaled from Gen1 Table 2-2)
- MRM modulator physics: transform-limited spectrum, chirp, bandwidth
- Chromatic dispersion penalty at 1310 nm operating band
- Extinction ratio and OMA power analysis
- RIN noise penalty
- TDEC budget (modulator + driver + dispersion + ER)
- TX jitter budget (Dual Dirac, BER = 1×10^{-12})
- TX optical power chain: ELS → MRM → TP2
- ELS interface requirements

2. TX System Architecture



Measurement planes:

- **TP1:** ELS output (laser facet, before PMF)
- **TP2:** OCI chiplet fiber output (OCI optical TX compliance)
- **TP3:** Receiver fiber input (after 500 m fiber)

3. TX Driver Electrical Specification

The OCI Gen2 MRM driver is co-designed and co-integrated with the MRM on the GF photonics platform. This integration changes the driver spec relative to a standard SerDes TX driver. The following taxonomy follows industry convention for NRZ drivers (reference: A. Izadi, Lightmatter internal communication).

3.1 Differential Output Swing

The driver must swing the MRM PN junction voltage from the "off-resonance" (high transmission = "1") state to "on-resonance" (low transmission = "0") state.

Parameter	Value	Notes
Vswing, differential, min	0.5 Vpp	Minimum for ER ≥ 3.5 dB
Vswing, differential, typ	1.0 Vpp	Design target for ER ≈ 4.5 dB

Parameter	Value	Notes
Vswing, differential, max	1.5 Vpp	Limited by GF device reliability
Common-mode output	Device-dependent	Set by MRM DC bias for resonance alignment

Higher swing increases ER but increases driver power consumption quadratically. The operating point balances ER vs power and ring heating (resonance shift from self-heating).

3.2 Output Impedance and Return Loss

NOT REQUIRED for integrated MRM driver.

The MRM electrode is a lumped RC load driven by a co-located driver — there is no transmission line between them. The standard 50Ω output impedance and return-loss specs (ERL, common-mode return loss) that appear in external-drive TX specs do not apply here.

The relevant electrical load seen by the driver is:

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C_MRM ≈ 10–30 fF      (depletion PN junction)
R_series ≈ 10–30 Ω     (contact + via resistance)
τ_RC = R × C ≈ 0.3–0.9 ps → BW_RC ≈ 180–530 GHz (not the bandwidth limiter)
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The MRM optical photon lifetime (BW_ring ≈ 60 GHz target) is the dominant bandwidth constraint — not the driver RC product.

3.3 TX Equalization

From Arash Izadi: *"Transmitter equalization coefficient range and resolution (ex. Pre, Post, 3 bit, 0.3x max)"*

The driver implements a 3-tap FIR equalizer to pre-compensate ISI introduced by the combined driver + MRM bandwidth channel:

$$y(k) = c_{-1} \cdot x(k+1) + c_0 \cdot x(k) + c_{+1} \cdot x(k-1)$$

where $x(k) \in \{-1, +1\}$ (NRZ data symbols)

Tap	Name	Range	Resolution	Typical value	Purpose
c ₋₁	Pre-cursor	0 to -0.3	3-bit (8 steps)	-0.10	Pre-cursor ISI from driver latency/phase
c ₀	Main cursor	remainder	—	~0.70	Signal amplitude (normalized)

Tap	Name	Range	Resolution	Typical value	Purpose
c_{+1}	Post-cursor	0 to -0.3	3-bit (8 steps)	-0.22	Post-cursor ISI from MRM Lorentzian rolloff

Normalization constraint: $|c_{-1}| + c_0 + |c_{+1}| = 1$ (total power preserved)

With $c_{-1} = -0.10$ and $c_{+1} = -0.22$: $c_0 = 1 - 0.10 - 0.22 = 0.68$

The post-cursor is dominant because the MRM's Lorentzian transfer function creates significant post-cursor ISI at 106G ($BW_MRM/BR = 60/106 \approx 0.57$ — well below Nyquist).

Step size: With 3-bit resolution over a 0.3 range: $step = 0.3/7 \approx 0.043$ per tap (0 to 0.3 in 8 levels). This gives sufficient resolution to minimize residual ISI.

3.4 Rise/Fall Time Constraints

Rise and fall time (20%–80%) at the MRM electrode sets the final electrical slew rate before optical conversion. The MRM's photon lifetime then further limits the optical edge rate.

Parameter	Spec	Notes
Rise time, max (20-80%)	$\leq 6\text{ ps}$	At MRM electrode, after TX EQ
Fall time, max (20-80%)	$\leq 6\text{ ps}$	Symmetric target before asymmetric correction
Rise time, driver BW implied	$BW \geq 58\text{ GHz}$	$0.35 / 6\text{ ps}$
Optical rise time at TP2	$\leq 8.5\text{ ps}$	After MRM conversion (OCI optical spec)

The electrical 6 ps target at the driver output gives margin for the MRM optical bandwidth to further limit the optical edge to $\leq 8.5\text{ ps}$ at TP2.

3.5 Asymmetrical Peaking and Rise/Fall Time Control

Lightmatter-specific feature (per A. Izadi): *"To cancel MRM nonlinearity."*

The MRM resonance has a Lorentzian lineshape, making the transmission a nonlinear function of the drive voltage. The CW laser is parked slightly off-resonance (typically blue-detuned). This creates asymmetric optical edge speeds:

Optical transfer: $T(V) = T_{\text{max}} / (1 + (V - V_{\text{res}})^2 / (\Delta V_{\text{FWHM}}/2)^2)$

Near resonance ($V \approx V_{\text{res}}$): steep slope → fast optical transition (one polarity)

Far from resonance: shallow slope → slow optical transition (other polarity)

Result at TP2 (without correction):

- One edge (into resonance): fast optical rise → short transition
- Other edge (out of resonance): slow optical rise → long transition
- Consequence: **asymmetric eye, increased DCD, TDEC contribution**

Correction (driver-applied asymmetric peaking):

- Pre-emphasis amplitude asymmetry: apply more voltage overshoot on the slow edge
- Edge rate asymmetry: tune rising vs falling slew rate independently
- Target: $tr_{\text{optical}} \approx tf_{\text{optical}}$ at TP2, minimizing DCD

This requires characterization of the specific MRM bias point and resonance shape, and calibration per-channel (each MRM ring may have a slightly different Lorentzian).

3.6 Jitter Limits

Per A. Izadi: *"Random and deterministic jitter limits (DJ, RJ, PSIJ)"*

At BER = 1×10^{-12} (Q = 7.035, 2Q = 14.07):

$$TJ(1e-12) = DJ_{pp} + 14.07 \times \sigma_{RJ}$$

Jitter type	Symbol	Allocation	UI (9.41 ps)	Source
Duty cycle distortion	DCD	$\leq 0.40 \text{ ps}_{pp}$	$\leq 4.3\%$	Clock divider tr/tf mismatch
ISI — electrical	ISI_elec	$\leq 0.60 \text{ ps}_{pp}$	$\leq 6.4\%$	Residual after TX EQ
ISI — optical (MRM)	ISI_opt	$\leq 0.40 \text{ ps}_{pp}$	$\leq 4.3\%$	MRM BW + CD combined
Bounded uncorrelated	BUJ	$\leq 0.30 \text{ ps}_{pp}$	$\leq 3.2\%$	Crosstalk from adjacent rings
Periodic sinusoidal	PSIJ	$\leq 0.30 \text{ ps}_{pp}$	$\leq 3.2\%$	VCO/supply noise, PLL ref feedthrough
Total DJ_{pp}		$\leq 2.00 \text{ ps}_{pp}$	$\leq 21.3\% \text{ UI}$	RSS of bounded terms
Random jitter	σ_{RJ}	$\leq 0.15 \text{ ps}_{rms}$	$\leq 1.6\% \text{ UI}$	SerDes TX PLL VCO
Total TJ at BER=1e-12		$\leq 4.11 \text{ ps}_{pp}$	$\leq 43.7\% \text{ UI}$	$= 2.00 + 14.07 \times 0.15$
Eye opening		$\geq 5.30 \text{ ps}$	$\geq 56.3\% \text{ UI}$	$= 9.41 - 4.11$

3.7 Eye Mask

The eye mask combines all the above into a single go/no-go compliance test. Eye mask compliance is measured at the TX output (electrical at the MRM electrode, optical at TP2).

Electrical eye mask (100GAUI-1, IEEE 802.3ck Annex 120F):

Normalized to UI = 9.41 ps. The mask defines a forbidden zone centered in the eye. Typical inner mask corners:

$$\begin{aligned}(x_1, \pm y_1) &= (0.25 \text{ UI}, \pm 0.6 \times \text{amplitude_pp}) \\ (x_2, \pm y_2) &= (0.40 \text{ UI}, \pm 0.3 \times \text{amplitude_pp})\end{aligned}$$

No waveform sample shall fall within the mask inner region at the specified BER = 1e-12.

Optical eye mask (OCI Gen2 at TP2):

Defined by the OCI Gen2 specification (TBD at time of writing; expected to follow OCI Gen1 eye mask scaled to 106.25 GBaud). Key constraints embedded in optical spec: TDEC $\leq$ 3.4 dB, transition time $\leq$ 8.5 ps, overshoot $\leq$ 22%, ER $\geq$ 3.5 dB.

4. Gen1 → Gen2 Scaling Rules

Parameter	Gen1	Gen2	Scale
Baud rate	53.125 GBaud	106.25 GBaud	×2
UI	18.82 ps	9.41 ps	÷2
TDEC ref RX BW	26.5625 GHz	53.125 GHz	×2
Rise time spec (20-80%), optical	17 ps	~8.5 ps	÷2
Rise time spec (20-80%), driver	~10 ps	~6 ps	÷2
Spectral BW (transform-limited)	~0.30 nm	~0.61 nm	×2
Optical power specs (OMA, Pavg)	—	unchanged	×1
TDEC limit	3.4 dB	3.4 dB	×1
ER	3.5 dB min	3.5 dB min	×1
RIN_OMA	-138 dB/Hz	-138 dB/Hz	×1
CD (fiber channel)	-0.9 to +1.7 ps/nm	same	×1
Driver jitter BER	1×10 ⁻¹²	1×10⁻¹²	×1
Q factor (driver jitter BER=1e-12)	7.035	7.035	×1
TDEC BER (optical compliance)	2.4×10 ⁻⁴	2.4×10⁻⁴	×1

Note on TDEC tightness: The Gen2 reference receiver has twice the bandwidth, so it passes more of the MRM and driver roll-off into the eye measurement. A MRM with Gen1 bandwidth (~30 GHz) would generate significantly more ISI and fail TDEC at Gen2. The MRM and driver must both target $\geq$ 60 GHz 3dB bandwidth.

5. OCI Gen2 TX Optical Specifications

Derived from OCI 200G v1.0 Table 2-2 with Gen2 scaling.

Wavelengths: Group A: 1307.8–1315.1 nm; Group B: 1327.5–1335.0 nm (4 channels per group).

5.1 TX Compliance Spec Table (at TP2)

Parameter	Gen1 (53G)	Gen2 (106G)	Unit	Notes
Baud rate	53.125	106.25	GBaud	
SMSR	≥ 30	≥ 30	dB	Laser side-mode suppression
Pavg per channel, min	−8.5	−8.5	dBm	
Pavg per channel, max	0	0	dBm	
Pavg total per group (4λ), max	+6	+6	dBm	Sum of 4 channels
OMA per channel, min	max(−5.5, −6.9+TDEC)	same formula	dBm	TDEC in dB
OMA per channel, max	−1	−1	dBm	
TDEC, max	3.4	3.4	dB	Ref RX: BT-4, BW = BR/2 = 53.125 GHz
dTDEC SSPR-PRBS13 , max	0.4	0.4	dB	Pattern sensitivity
ER (extinction ratio), min	3.5	3.5	dB	P1/P0
ER typical	4.5	4.5	dB	Design target
Transition time (20-80%), max	17	~8.5	ps	Optical, at TP2
Overshoot / undershoot, max	22	22	%	Relative to OMA
RIN_OMA, max	−138	−138	dB/Hz	At ORL = 21.4 dB
Squelched OMA, max	−15	−15	dBm	Per channel, during squelch
ORL tolerance	21.4	21.4	dB	
TX data path reflectance	−19	−19	dB	Back-reflection into fiber
OE laser input reflectance	−26	−26	dB	

5.2 OMA Compliance Floor vs TDEC

$$\text{OMA}_{\min} = \max(-5.5 \text{ dBm}, -6.9 \text{ dBm} + \text{TDEC}_{\text{dB}})$$

TDEC (dB)	OMA _{min} (dBm)	Governing term
0.0	-5.5	Fixed floor
1.4	-5.5	Crossover point
1.9 (design target)	-5.0	TDEC-dependent
3.4 (spec max)	-3.5	TDEC-dependent

6. MRM Modulator Physics

6.1 Transform-Limited Spectrum ($\alpha = 0$)

The MRM modulates intensity by shifting the ring resonance on/off the CW carrier wavelength. Unlike a DML, the laser frequency is fixed — only the amplitude changes. This gives:

- **Chirp parameter $\alpha = 0$** (no frequency modulation)
- **Transform-limited optical spectrum** (narrowest possible for a given data rate)
- Sinc²-shaped power spectrum with 3 dB bandwidth $\approx$ baud rate B

Contrast with directly-modulated transmitters (Lecture 7):

Transmitter type	α	Spectral width	Dispersion reach
DML	4–6	$\Delta\lambda \times \sqrt{(\alpha^2+1)} \approx 4\text{--}6\times$ wider	Limited; 4× shorter than EML
EML (MZM/EAM)	≈ 0	$\Delta\lambda_{\text{TL}} = \lambda^2 B / c$	Maximum reach
MRM (OCI Gen2)	≈ 0	$\Delta\lambda_{\text{TL}}$	Maximum reach

6.2 MRM Bandwidth Requirement

The MRM optical 3dB bandwidth (set by the ring photon lifetime) must be large enough to support 106.25 GBaud NRZ with acceptable TDEC. Ring photon lifetime:

$$\tau_{\text{ph}} = 1 / (2\pi \times \text{BW}_{\text{MRM}}) \quad \rightarrow \quad \text{BW}_{\text{MRM}} = 60 \text{ GHz}: \tau_{\text{ph}} = 2.65 \text{ ps}$$

The Lorentzian roll-off creates post-cursor ISI. The faster the ring (shorter τ_{ph} , wider BW), the lower the ISI but the higher the required coupling ($\rightarrow$ higher IL trade-off). See §10.4 for the BW vs TDEC trade-off table.

6.3 MRM Insertion Loss

- On-state (through-port when off-resonance): **$\sim 2\text{--}3 \text{ dB}$** (waveguide + coupler)
- Off-state extinction (when on-resonance): sets the ER floor
- Thermal sensitivity: $\sim 80 \text{ pm}/^\circ\text{C} \rightarrow$ heater closed-loop required

- WDM: 4 MRM rings per group, individual thermal tuning per ring

7. Spectral Linewidth and Chromatic Dispersion

7.1 Transform-Limited Spectral Linewidth

$$\Delta\lambda = (\lambda^2 / c) \times B$$

[from Lecture 7]

At $\lambda = 1311 \text{ nm}$ (Group A center), $B = 106.25 \text{ GHz}$:
 $\Delta\lambda = (1311 \times 10^{-9})^2 \times 106.25 \times 10^9 / (3 \times 10^8) = 0.609 \text{ nm}$

Quantity	Gen1	Gen2
Baud rate	53.125 GHz	106.25 GHz
$\Delta\lambda$ (transform-limited)	0.305 nm	0.609 nm
ELS laser linewidth	$\leq 0.0007 \text{ nm}$ (1 MHz)	same
Dominant contributor	Signal modulation	Signal modulation

The ELS linewidth (0.0007 nm) is negligible vs the modulation spectral width (0.609 nm). The system is firmly in the transform-limited (not source-limited) regime.

7.2 Chromatic Dispersion Penalty

OCI Gen2 operates at 1308–1335 nm, near SMF-28 zero dispersion ($\lambda_0 \approx 1310 \text{ nm}$):

Channel group	Center λ	Typical D	500 m CD
A	1311 nm	$\sim 0.1 \text{ ps}/(\text{nm} \cdot \text{km})$	$\sim 0.05 \text{ ps}/\text{nm}$
B	1331 nm	$\sim 0.7 \text{ ps}/(\text{nm} \cdot \text{km})$	$\sim 0.35 \text{ ps}/\text{nm}$

OCI spec maximum total channel dispersion: $|CD| \leq 1.7 \text{ ps}/\text{nm}$

Maximum pulse broadening (worst case, Group B at spec limit):

$$\Delta T = |CD_{\text{max}}| \times \Delta\lambda = 1.7 \text{ ps}/\text{nm} \times 0.609 \text{ nm} = 1.04 \text{ ps} = 0.11 \text{ UI}$$

1 dB dispersion penalty condition (Lecture 7): $\Delta T \leq UI/2 = 4.7 \text{ ps}$

At $\Delta T = 1.04 \text{ ps} \rightarrow$ **well below 1 dB threshold. CD penalty < 0.1 dB.**

This is why OCI Gen2 targets 1310 nm: near-zero dispersion eliminates CD as a significant TX budget line item. A 1550 nm design at the same baud rate would incur:

$$D_{\text{SMF}}(1550\text{nm}) \approx 17 \text{ ps}/(\text{nm} \cdot \text{km})$$

$$\Delta T = 17 \times 0.5 \times 0.609 \text{ nm} \approx 5.2 \text{ ps} = 0.55 \text{ UI} \rightarrow \sim 0.8 \text{ dB TDEC}$$

8. Extinction Ratio and OMA Analysis

8.1 Definitions

$$\begin{aligned} \text{ER} &= P_1 / P_0 && (\text{linear ratio}) \\ \text{OMA} &= P_1 - P_0 \\ P_{\text{avg}} &= (P_1 + P_0) / 2 \\ \text{OMA} &= 2(\text{ER}-1)/(\text{ER}+1) \times P_{\text{avg}} \end{aligned}$$

8.2 Average-Power Receiver Penalty

For a TIA-noise-limited receiver (constant noise), sensitivity referenced to P_{avg} :

$$\text{PP}_{\text{avg}} = (\text{ER} + 1) / (\text{ER} - 1) \quad [\text{linear, Lecture 7}]$$

ER (dB)	ER (linear)	PP_avg (dB)	OMA/P_avg
3.5 (min spec)	2.24	4.18 dB	0.765 (-1.16 dB)
4.5 (typical)	2.82	3.22 dB	0.952 (-0.21 dB)
6.0	3.98	2.09 dB	0.992 (-0.04 dB)
∞ (ideal NRZ)	∞	0 dB	2 (+3.01 dB)

8.3 OMA-Based Specification Advantage

The OCI spec is **OMA-based**. At constant OMA, ER incurs no receiver noise penalty (the TIA noise floor is fixed; OMA sets the SNR directly). ER still matters for:

1. P_0 leakage $\rightarrow$ reduced eye amplitude $\rightarrow$ TDEC contribution (§10.2)
2. P_{avg} budget $\rightarrow$ MRM self-heating, group power limit
3. Higher ER $\rightarrow$ lower required P_{avg} for same OMA $\rightarrow$ saves ~ 1 dB ELS power (§12.2)

8.4 P_0 Leakage $\rightarrow$ Eye Amplitude Reduction (TDEC Contribution)

$$\text{Normalized eye amplitude} = 1 - 2/(\text{ER}+1) = (\text{ER}-1)/(\text{ER}+1)$$

$$\text{At ER} = 3.5 \text{ dB (2.24): normalized} = 1.24/3.24 = 0.383 \rightarrow \text{eye loss} \approx 0.4 \text{ dB TDEC}$$

At ER = 4.5 dB (2.82): normalized = $1.82/3.82 = 0.476 \rightarrow$ eye loss ≈ 0.3 dB TDEC

9. RIN Budget

9.1 RIN-to-SNR Calculation

$SNR_{RIN} = 1 / (RIN_{OMA} \times BW_{noise})$
 $BW_{noise} \approx 0.75 \times BR = 0.75 \times 106.25 \text{ GHz} = 79.7 \text{ GHz}$ (BT-4 effective noise BW)

$RIN_{OMA} = -138 \text{ dB/Hz} = 1.585 \times 10^{-14} \text{ /Hz}$ (TX compliance spec max)

$SNR_{RIN} = 1 / (1.585 \times 10^{-14} \times 79.7 \times 10^9) = 1 / 1.263 \times 10^{-3} = 792 \rightarrow 28.99 \text{ dB}$

9.2 RIN Penalty at BER = 1×10^{-12}

$Q = 7.035, Q^2 = 49.5 \rightarrow$ required SNR = 49.5 $\rightarrow 16.94 \text{ dB}$

RIN margin = $28.99 - 16.94 = 12.05 \text{ dB}$

RIN power penalty = $10 \log_{10}(1 + Q^2/SNR_{RIN}) = 10 \log_{10}(1 + 49.5/792) = 10 \log_{10}(1.0625)$
 $\approx 0.26 \text{ dB}$ (small, included in TDEC margin)

At the optical TDEC measurement BER = 2.4×10^{-4} ($Q = 3.54, Q^2 = 12.5$):

RIN margin = $28.99 - 10.98 = 18.0 \text{ dB} \rightarrow$ RIN penalty = 0.07 dB (negligible)

10. TDEC Budget

10.1 TDEC Definition and Measurement

TDEC measures optical eye closure relative to an ideal reference at the OCI TX compliance BER threshold (note: this is NOT the driver jitter BER):

$TDEC = -10 \log_{10}(OMA_{meas} / OMA_{ref})$

Gen2 measurement conditions:

- Reference receiver: 4th-order Bessel-Thomson, BW = **53.125 GHz** (= BR/2)

- Equalizer: **none** at the RX
- Test pattern: **SSPR** (Single Spectrum Pseudo-Random)
- Eye histogram center: $0.4 \text{ UI} \times 9.41 \text{ ps} = \mathbf{3.76 \text{ ps}}$ from each edge
- BER threshold: $\mathbf{2.4 \times 10^{-4}}$ (KP4 pre-FEC optical compliance)
- dTDEC: $|\text{TDEC_SSPR} - \text{TDEC_PRBS13}| \leq 0.4 \text{ dB}$

10.2 TDEC Contributors — Budget Allocation

Contributor	Mechanism	Penalty (dB)	Notes
MRM optical BW	Lorentzian rolloff at 60 GHz → post-cursor ISI	0.6	BW_MRM = 60 GHz target
TX driver + EQ residual	CDNS L250 SerDes TX FFE; residual ISI through 53 GHz ref RX	0.5	Post 3-tap EQ
Finite extinction ratio	P0 leakage at ER = 4.5 dB	0.3	See §8.4
MRM nonlinearity	Lorentzian → asymmetric eye; partially corrected by asymmetric driver	0.2	Residual after driver correction
DCD / rise-fall asymmetry	tr ≠ tf optical, after asymmetric peaking correction	0.15	Residual DCD
Chromatic dispersion	1.04 ps broadening × 53 GHz ref RX ISI	0.1	Near-zero at 1310 nm
Overshoot / ringing	MRM coupled-ring response; driver pre-emphasis	0.15	Residual
Design sub-total		2.0	
Margin to spec		1.4	41% headroom
Spec limit (max)		3.4	OCI Gen2 optical requirement

10.3 MRM Bandwidth vs TDEC

BW_MRM	BW_MRM / BR	TDEC_MRM contribution	Driver EQ needed
40 GHz	0.38	~1.2 dB	Aggressive 3-tap
60 GHz	0.56	~0.6 dB	2-tap sufficient
80 GHz	0.75	~0.3 dB	1-tap sufficient
106 GHz	1.0	~0.1 dB	No EQ needed

Target: BW_MRM ≥ 60 GHz → TDEC_MRM ≤ 0.6 dB.

11. TX Jitter Budget — Dual Dirac Model

BER = 1×10⁻¹² (Q = 7.035, 2Q = 14.07) — applies to TX driver electrical compliance.

11.1 Jitter Taxonomy (Lecture 10, Palermo ECEN720)

$$TJ(t) = RJ(t) * DJ(t) \quad (\text{PDF convolution})$$
$$DJ(t) = SJ/PSIJ(t) * DCD(t) * ISI(t) * BUJ(t)$$

Type	Symbol	Source	Statistics
Random Jitter	RJ	TX PLL VCO thermal noise	Gaussian (unbounded); σ_{RJ}
Periodic / Sinusoidal	PSIJ	Power supply noise, PLL ref feedthrough	Periodic; arcsine PDF; bounded
Duty Cycle Distortion	DCD	Clock divider tr/tf mismatch, asymmetric buffers	Dual delta; bounded
ISI Jitter	ISI	BW-limited channel (electrical + optical)	Bounded; reduced by TX EQ
Bounded Uncorrelated	BUJ	Crosstalk from adjacent WDM rings	Bounded; uncorrelated to victim

11.2 Dual Dirac Model

$$DJ(t) = \frac{1}{2} \delta(t - DJ_{pp}/2) + \frac{1}{2} \delta(t + DJ_{pp}/2)$$
$$TJ(BER) = DJ_{pp} + 2 \times Q(BER) \times \sigma_{RJ}$$

At BER = 1×10⁻¹²: Q = 7.035

$$TJ(1e-12) = DJ_{pp} + 14.07 \times \sigma_{RJ}$$

11.3 Gen2 TX Driver Jitter Budget

UI_Gen2 = 9.41 ps, BER target = 1×10⁻¹²

Component	Budget	% UI	Governed by
DCD	0.40 ps _{pp}	4.3%	SerDes TX clock divider duty cycle
ISI — electrical (after EQ)	0.60 ps _{pp}	6.4%	Residual ISI through 3-tap TX FFE
ISI — optical (MRM 60 GHz)	0.40 ps _{pp}	4.3%	Ring photon lifetime + CD = 1 ps
BUJ (adjacent ring crosstalk)	0.30 ps _{pp}	3.2%	4λ WDM rings on same die, thermal/optical
PSIJ (VCO supply, PLL ref)	0.30 ps _{pp}	3.2%	LDO-regulated supply; spread-spectrum

Component	Budget	% UI	Governed by
DJ_pp total	2.00 ps_pp	21.3% UI	Sum of bounded components
σ_{RJ}	0.15 ps_rms	1.6% UI	SerDes TX PLL VCO, ~100 fs/√Hz integrated
TJ at BER = 1e-12	4.11 ps_pp	43.7% UI	= 2.00 + 14.07×0.15
Eye opening (timing)	5.30 ps	56.3% UI	= 9.41 – 4.11

11.4 Jitter–TDEC Relationship

TDEC captures **vertical** eye closure (ISI-amplitude); jitter captures **horizontal** eye closure (timing). The TDEC measurement excludes the outer 0.4 UI per edge (histogram centering), providing tolerance for horizontal jitter up to:

$$\begin{aligned} \text{DJ_pp allowance within TDEC window} &= 2 \times 0.4 \text{ UI} - 2Q_{\text{opt}} \times \sigma_{RJ} \\ (\text{where } Q_{\text{opt}} &= 3.54 \text{ at optical BER} = 2.4e-4) \\ &= 2 \times 3.76 \text{ ps} - 7.07 \times 0.15 \text{ ps} = 7.52 - 1.06 = 6.46 \text{ ps_pp} \end{aligned}$$

$$\text{Design DJ_pp} = 2.00 \text{ ps_pp} \rightarrow \text{TDEC window margin} = 6.46 - 2.00 = 4.46 \text{ ps_pp} \checkmark$$

12. TX Optical Power Chain Budget

12.1 Per-Channel Loss (ELS → TP2)

Element	Nominal IL	Range	Notes
PMF pigtail + connectors	0.8 dB	0.5–1.0	APC connector, 0.5 m PMF
OCI chiplet fiber coupler (in)	0.7 dB	0.5–0.8	Lensed fiber or grating coupler
MRM on-state insertion loss	2.5 dB	2.0–3.5	Waveguide + coupler through-port
On-chip waveguide routing	0.7 dB	0.5–1.0	Bends, crossings, tapers
Fiber attach (out)	0.3 dB	0.3–0.5	SMF coupling
Total TX path loss	5.0 dB	3.8–6.8	

12.2 ELS Power Requirement per Channel

Design target: TDEC = 2.0 dB → OMA_{TP2} = –5.0 dBm (from §5.2)

$$\text{At } ER = 4.5 \text{ dB (2.82): } OMA = 0.952 \times P_{\text{avg}} \rightarrow P_{\text{avg}} = OMA/0.952$$

$$P_{\text{avg}} \text{ at TP2} = -5.0 \text{ dBm} + 0.21 \text{ dB} = -4.79 \text{ dBm}$$

$$\text{Required ELS CW power per channel} = -4.79 + 5.0 = +0.21 \text{ dBm (1.05 mW)}$$

Higher ER improves ELS efficiency: at ER = 6.0 dB (3.98 linear):

$$\text{OMA} = 0.992 \times P_{\text{avg}} \rightarrow P_{\text{avg}} \approx \text{OMA} \rightarrow \text{Saved } \sim 1 \text{ dB vs ER} = 3.5 \text{ dB}$$

12.3 Multi-Channel Power Summary

Condition	OMA/ch (TP2)	Pavg/ch (TP2)	ELS/ch	4-ch ELS total
TDEC = 2.0 dB (design)	-5.0 dBm	-4.8 dBm	+0.2 dBm (1.05 mW)	4.2 mW
TDEC = 3.4 dB (spec max)	-3.5 dBm	-3.3 dBm	+1.7 dBm (1.48 mW)	5.9 mW
OMA max (-1 dBm)	-1.0 dBm	-0.8 dBm	+4.2 dBm (2.63 mW)	10.5 mW

Group power limit: $4 \times \text{max_OMA} = 4 \times 1.0 \text{ mW} = 4.0 \text{ mW} \rightarrow \mathbf{+6.0 \text{ dBm} \leq 6 \text{ dBm spec } \checkmark}$ At design target: $4 \times 0.32 \text{ mW} = 1.28 \text{ mW} \rightarrow \mathbf{-1.0 \text{ dBm, well within limit.}}$

12.4 Fiber Link Budget (TP2 → TP3)

Parameter	Value	Source
Fiber length	500 m	OCI Gen2 reach spec
SMF-28 attenuation at 1310 nm	0.35 dB/km	Typical
Fiber loss (500 m)	0.18 dB	
Connector at TP3	0.5 dB	1 connector
Other (splices, bends)	< 0.3 dB	
Total fiber IL (TP2 → TP3)	≤ 2.5 dB	OCI Gen2 spec max

At design-target OMA_TP2 = -5.0 dBm:

$$\text{OMA_TP3} = -5.0 - 2.5 = -7.5 \text{ dBm}$$

This feeds into [106G_NRZ_CPO_RX_Budget.md](#) as the OMA input to the TIA sensitivity analysis.

13. ELS Requirements

From OCI 200G v1.0 Table 2-6, applicable to Gen2:

Parameter	Spec	Notes
Laser RIN	≤ -144 dB/Hz	At max ORL; 6 dB tighter than TX spec
Laser linewidth	≤ 1 MHz (≈ 0.7 pm)	Negligible vs 609 pm modulation width
SMSR	≥ 30 dB	

Parameter	Spec	Notes
Polarization extinction ratio	≥ 16 dB	TE-only MRM modulation
ELS output reflectance	≤ -26 dB	Back-reflection into laser
ORL tolerance	≤ -26 dB	ELS must tolerate reflections from fiber plant
Output power	IS	Implementation-specific; see §12.3

Polarization: MRM is TE-only. PMF from ELS to OCI chiplet is mandatory. Polarization rotation $> (30^\circ$ from TE) costs > 0.5 dB in effective OMA — a hidden loss item.

CW stability: ± 0.5 dB ELS power drift $\rightarrow \pm 0.5$ dB OMA drift, consuming TDEC margin. Closed-loop monitoring (per-channel optical power tap + feedback to ELS gain or VOA) is strongly recommended.

14. Standards Compliance Context

14.1 Electrical Interface: IEEE 802.3ck-2022

IEEE 802.3ck governs the 100GAUI-1 chip-to-chip and chip-to-module electrical interface between the CDNS L250 SerDes and the MRM driver (or package boundary):

802.3ck Parameter	Spec	Clause
Signaling rate	106.25 GBd $\pm$ 100 ppm	162.9
TX output ERL	≥ 10 dB	162.9.4.8
TX output SNDR	≥ 26 dB	162.9.4.6
TX DJ _{pp}	≤ 0.20 UI = 1.88 ps	162.9.4.7
TX RJ _{rms}	≤ 0.010 UI = 0.094 ps	162.9.4.7
TX equalization	Pre, Post, 3-bit, $\pm 0.3\times$ max	120F.3.1.5

Note: The integrated MRM driver is exempt from the ERL/impedance specs (no package launch). The 802.3ck electrical TX jitter specs apply at the SerDes output pad (before the MRM driver redrives). The MRM driver adds its own jitter contribution tracked in §11.3.

14.2 Optical: OCI Gen2 Spec (derived from 200G-OCI v1.0)

Primary compliance is the OCI Gen2 optical spec at TP2 (§5). The IEEE 802.3ck covers only the electrical interface; there is no 802.3ck optical PMD for this wavelength/topology. The relevant draft for future 200G optical short-reach (P802.3dj) covers electrical and optical interfaces at 200G–1.6T but does not directly specify the WDM-MRM CPO topology used in OCI Gen2.

14.3 FEC Context

FEC	Pre-FEC BER	Q	Post-FEC BER	Use
None (raw link)	—	7.035	1×10^{-12}	Driver jitter spec

FEC	Pre-FEC BER	Q	Post-FEC BER	Use
KP4 RS(544,514)	2.4×10^{-4}	3.54	$< 10^{-13}$	OCI TDEC optical spec
RS(255,239)	10^{-3}	3.09	$< 10^{-12}$	Alternative (not OCI primary)

15. TX Budget Summary

Metric	Spec / Target	Design Value	Margin	Risk
TDEC	≤ 3.4 dB	2.0 dB	1.4 dB	Medium — MRM BW critical
dTDEC	≤ 0.4 dB	0.2 dB	0.2 dB	Low
OMA_min (at TDEC=2.0)	≥ -5.0 dBm	-5.0 dBm	0 dB	High — tight with ELS IL
OMA_max	≤ -1.0 dBm	-5.0 dBm	4.0 dB	Low
ER	≥ 3.5 dB	4.5 dB	1.0 dB	Low
Optical tr/ta	≤ 8.5 ps	~7 ps	1.5 ps	Medium — MRM photon lifetime
Driver tr/ta	≤ 6 ps	target	TBD	Medium
Driver DJ_pp	≤ 2.00 ps (budget)	2.00 ps	0 ps	Medium — full allocation used
Driver σ_{RJ}	≤ 0.15 ps	0.15 ps	0 ps	Medium — full allocation used
TJ at 1e-12	≤ 4.11 ps (43.7% UI)	4.11 ps	56.3% UI open	Medium
TX EQ: post-cursor	$\leq 0.3\times$	~0.22	0.08	Low
TX EQ: pre-cursor	$\leq 0.3\times$	~0.10	0.20	Low
Overshoot	$\leq 22\%$	10%	12%	Low
RIN penalty (1e-12)	—	0.26 dB	~12 dB margin	Low
CD penalty	—	0.1 dB	~0.8 dB vs limit	Low
Group power (4ch)	$\leq +6$ dBm	+1.0 dBm	5 dB	Low

16. Design Trade-offs and Risks

16.1 MRM Bandwidth vs Insertion Loss

Higher ring BW → smaller ring or higher coupling → more IL. At 60 GHz target BW, the ring must be critically coupled with low round-trip loss. Options:

- **Segmented electrodes:** reduce series resistance → higher RC BW
- **Travelling-wave design:** removes lumped-element RC limit
- **Larger ring radius:** lower optical loss but narrower FSR (WDM crosstalk risk)

16.2 TX EQ Coefficient Trade-off

Higher post-cursor coefficient reduces ISI but reduces main-cursor amplitude, lowering OMA for the same drive swing. At $c_{+1} = -0.22$:

$OMA_{after_EQ} = OMA_{no_EQ} \times (c_0 - |c_{+1}|) / 1$ = not a simple subtraction
(EQ redistributes energy; full Nyquist bandwidth analysis needed)

Measured TDEC vs EQ coefficient should be swept experimentally during bring-up.

16.3 Asymmetric Peaking Calibration

The MRM nonlinearity varies with:

- Bias point (heater-set wavelength offset from resonance)
- Temperature (resonance drift changes the nonlinearity slope)
- Drive amplitude (V_{swing} sets which portion of the Lorentzian is traversed)

A per-chip, per-temperature calibration table for the asymmetric pre-emphasis coefficients is likely needed. This adds firmware complexity but is necessary to minimize DCD at 106G.

16.4 Thermal Drift and Wavelength Lock

MRM resonance: ~ 80 pm/°C. WDM channel spacing: ~ 2.3 nm. Max tolerable drift before channel collision: $2300/80 = \mathbf{28.8^\circ C}$. On-chip thermal variation $\pm 10^\circ C$ + tile coupling $\pm 5^\circ C$ → heater loop settling < 50 ms. Heater power per ring: 1–5 mW. For 8 rings per chiplet (4 per group $\times$ 2 groups): **Heater power budget: 8–40 mW per chiplet.**

16.5 PSIJ from On-Chip Power Supply

At 106.25 GBaud, reference clock is 106.25/N GHz (with integer N SerDes divider). Power supply noise at multiples of this frequency appears as PSIJ. The SerDes PLL VCO supply must be LDO-regulated to keep $PSIJ_{pp} \leq 0.30$ ps. Switching noise from heater DAC updates should be isolated from the SerDes supply domain.

17. Next Steps

1. **MRM S21 measurement:** Measure modulation bandwidth from DC to 80 GHz. Target: 3dB BW ≥ 60 GHz. Record Lorentzian shape for asymmetric peaking design.

2. **Driver EQ optimization:** Sweep $[c_{-1}, c_{+1}]$ coefficients at 3-bit resolution. Minimize TDEC at TP2. Measure $dTDEC(SSPR \text{ vs } PRBS13) \leq 0.4 \text{ dB}$.
3. **Asymmetric peaking calibration:** Characterize MRM nonlinearity at design bias; iterate driver rise/fall asymmetry coefficients to minimize DCD in optical eye.
4. **TDEC test at TP2:** Drive at 106.25 GBaud SSPR; measure through 53.125 GHz BT-4 reference receiver; verify $TDEC \leq 3.4 \text{ dB}$ and $dTDEC \leq 0.4 \text{ dB}$.
5. **Electrical jitter measurement:** Extract DJ_{pp} , σ_{RJ} , $PSIJ_{pp}$ from bathtub curve at the SerDes TX output. Verify $TJ \leq 4.11 \text{ ps}$ at $BER = 1 \times 10^{-12}$.
6. **ER and OMA sweep:** Vary MRM bias (heater DAC) and drive swing (V_{swing}). Map ER vs OMA vs TDEC. Find optimal operating point.
7. **ELS power sweep:** Characterize OMA at TP2 vs ELS CW power. Confirm $\geq -5.0 \text{ dBm}$ OMA with 5 dB nominal TX path IL. Measure total group power.
8. **Thermal lock loop:** Validate heater feedback settling time $< 50 \text{ ms}$ over -10°C to $+85^\circ\text{C}$ range. Characterize $PSIJ$ from heater DAC switching.
9. **End-to-end BER:** OCI TX + 500 m SMF-28 + OCI RX. Measure BER vs OMA. Compare against RX budget sensitivity prediction.

Appendix A: Key Formulas

```
# Transform-limited spectral linewidth
 $\Delta\lambda = \lambda^2 \times B / c$ 

# ER average-power penalty
 $PP_{avg} = (ER + 1) / (ER - 1)$  [linear]

# OMA-Pavg relationship
 $OMA / P_{avg} = 2(ER - 1) / (ER + 1)$ 

# RIN SNR
 $SNR_{RIN} = 1 / (RIN_{linear} \times BW_{noise})$ 

# Dual Dirac total jitter
 $TJ(BER) = DJ_{pp} + 2 \times Q(BER) \times \sigma_{RJ}$ 

# Key Q values
 $Q(1 \times 10^{-12}) = 7.035 \rightarrow 2Q = 14.07$  [driver jitter spec]
 $Q(2.4 \times 10^{-4}) = 3.54 \rightarrow 2Q = 7.07$  [OCI TDEC optical compliance]

# TX FIR equalizer normalization
 $|c_{-1}| + c_0 + |c_{+1}| = 1$ 
 $c_0 = 1 - |c_{-1}| - |c_{+1}|$ 

# TDEC OMA floor
 $OMA_{min} = \max(-5.5 \text{ dBm}, -6.9 \text{ dBm} + TDEC_{dB})$ 
```

Chromatic dispersion pulse broadening
 $\Delta T = |\text{CD_total}| \times \Delta \lambda \quad [\text{ps_total} \times \text{nm} = \text{ps}]$

Appendix B: OCI Wavelength Plan (Gen2, same as Gen1)

Group	Channel	λ (nm)	D at λ	Notes
A	A0	1307.8	~0.1 ps/(nm·km)	
A	A1	1310.1	~0.05	Near λ_0
A	A2	1312.4	~0.1	
A	A3	1315.1	~0.15	
B	B0	1327.5	~0.6	
B	B1	1329.8	~0.7	
B	B2	1332.2	~0.7	
B	B3	1334.9	~0.8	

All channels within SMF-28 low-dispersion window. No WDM grid changes Gen1 → Gen2.